AI FOR MANUFACTURING

Day 1: Build Your AI HQ
Montcalm Community College

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WELCOME!

Get everything for today here:

Today's guide

github.com/grcc-resendiz/ai-mcc



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YOUR INSTRUCTORS

Jonnathan Resendiz

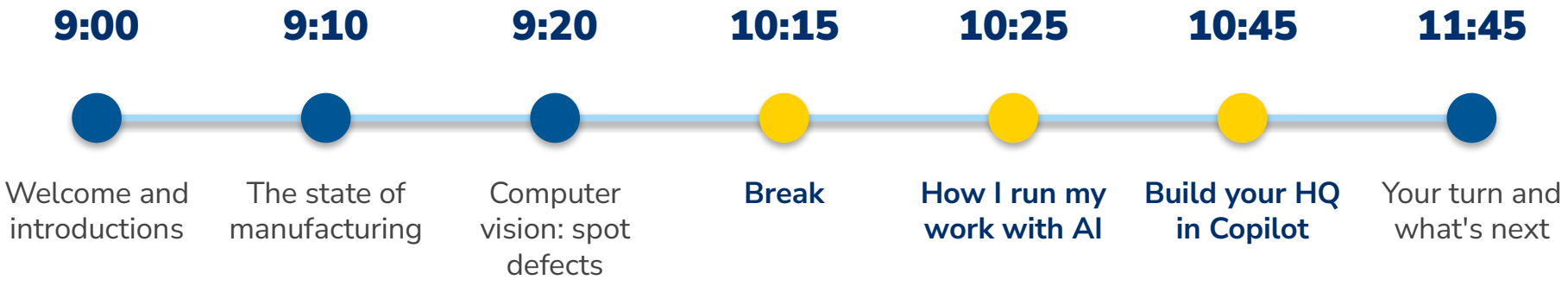
- ➔ Assistant Professor and Computer Information Systems Department Head, Grand Rapids Community College
- ➔ Faculty Director, GRCC AI Incubator
- ➔ M.S. in Computer Science, University of Texas at Dallas

Cheyenne Beach

- ➔ Automation and Advanced Industrial Technology Instructor, Montcalm Community College

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TODAY'S AGENDA



9:00 AM to 12:00 PM. Short talks, lots of hands-on. Your answers before 9:20 become what you build after the break.

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WHAT YOU'LL DO IN 3 DAYS

Each day builds on the one before.

DAY 1

Your AI HQ

Train an AI to spot defects, then build your own HQ in Copilot.

DAY 2

Skills and workflows

Turn your HQ into repeatable skills and workflows.

DAY 3

AI agents

Build agents and decide where they help most.

THE STATE OF MANUFACTURING

Let's start with you.

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WHAT KEEPS YOU UP AT NIGHT?

Shout them out. We'll come back to your answers when you build your HQ.

- Downtime
- Quality
- Scrap
- Hiring
- Training
- Scheduling
- Safety
- Yours?

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THE MANUFACTURING REALITY, 2024 TO 2033

3.8M

manufacturing jobs to fill by 2033

1.9M

of those could go unfilled

1 in 2

open jobs at risk of staying empty

Now

this is already happening, not a future problem

Source: Deloitte and The Manufacturing Institute, April 2024

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TWO BIG CHALLENGES

1

Workforce shortage

Not enough people to hire and train.

2

Technology adoption gap

The tools exist. Using them is the hard part.

We don't have a technology problem. **We have an adoption problem.**

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AI ADOPTION AS A SOLUTION

Why AI, and why now

- ➔ AI amplifies learning: use it to learn every other technology
- ➔ Invest in AI training, and you start solving both challenges
- ➔ No one is coming to save us. We build the solutions ourselves.

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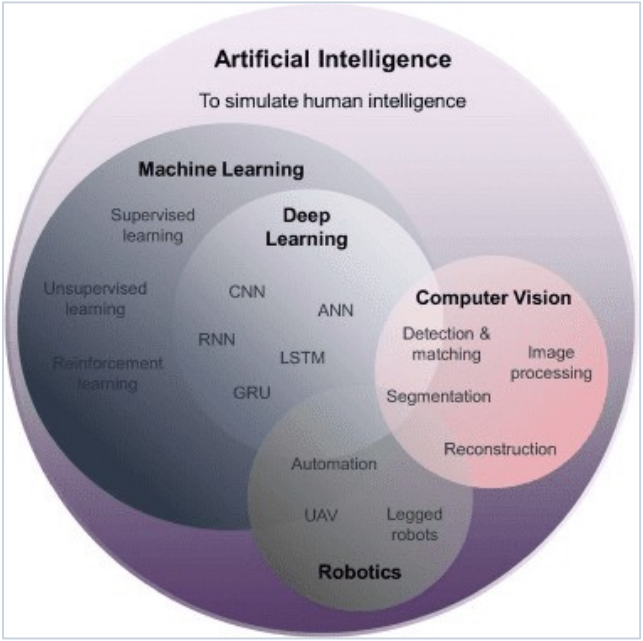
COMPUTER VISION

Let's watch AI learn from pictures.

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COMPUTER VISION: TEACHING COMPUTERS TO SEE

Computer vision is the part of AI that teaches computers to process and understand images and video.



Identify
Find the part in the picture

Classify
Decide what it is: OK or defective

Segment
Outline exactly where the flaw is

Today's lab is classification: OK or defective.

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COMPUTER VISION ON THE FLOOR



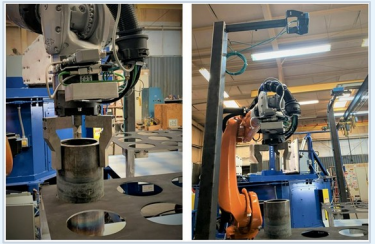
Quality inspection

Catch defects and parts that are out of spec.



Sorting

Sort parts by size, shape, color, or material.



Robot guidance

Guide pick and place and assembly.

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GREAT AT, STRUGGLES WITH

Know both before you buy anything.

Great at

- Consistent on every shift, no fatigue
- Faster decisions on the line
- Data on every part, not a sample
- Spotting safety hazards

Struggles with

- Changes in lighting and camera angle
- Subtle or rare defects
- Upfront cost for cameras and setup
- Privacy concerns from the floor

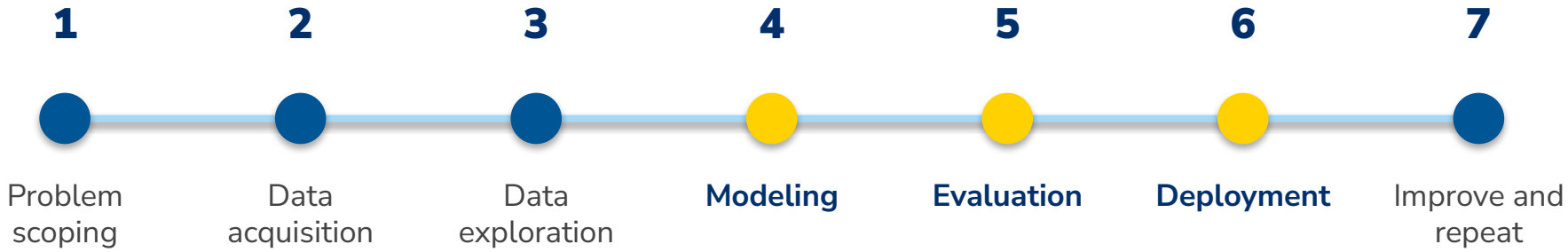
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HANDS-ON: SPOT THE DEFECT

Train your own model in 30 minutes.

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THE AI PROJECT CYCLE



Today's problem: is this cast part OK or defective? Each lab step is one stage of the cycle.

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STEP 1:
GET SET UP

Goal: a model that tells OK cast parts from defective ones.

DO

Download casting_data.zip from the guide and unzip it

DO

Go to teachablemachine.withgoogle.com

Get Started > Image Project > Standard image model

EXPECT

A folder of cast part pictures, already sorted into OK and defective

An empty project with Class 1 and Class 2

AI PROJECT CYCLE: PROBLEM SCOPING

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STEP 2: CLASS 1 IS “OK”

DO

Click the pencil icon and rename Class 1 to OK

DO

Click Upload and add the pictures from train/ok_front

AI PROJECT CYCLE: DATA ACQUISITION



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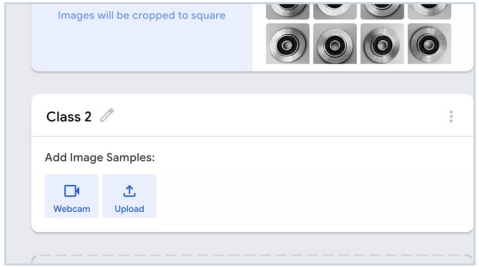
STEP 3: CLASS 2 IS “DEFECTIVE”

DO

Rename Class 2 to Defective

DO

Click Upload and add the pictures from train/def_front



AI PROJECT CYCLE: DATA ACQUISITION

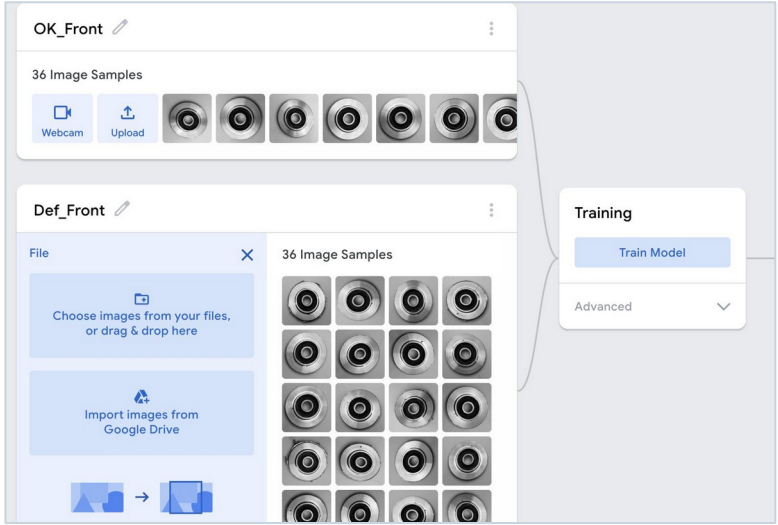
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STEP 4:
TRAIN IT

Keep the browser tab open while it trains.

DO

Click Train Model



AI PROJECT CYCLE: MODELING

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STEP 5: TEST IT ON NEW PARTS

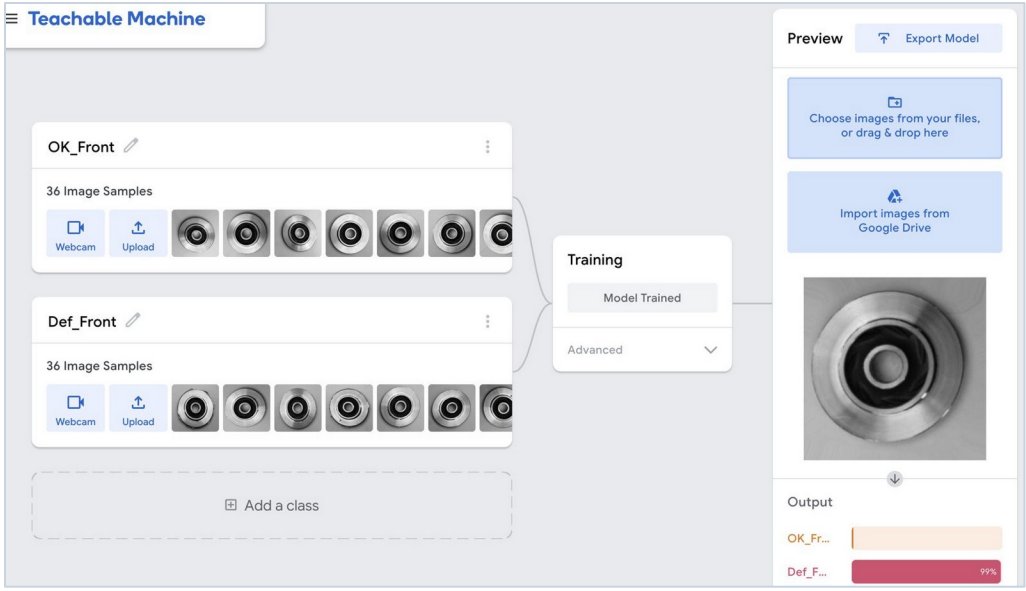
DO

In Preview, switch Webcam to File and upload pictures from the test folders

EXPECT

Most parts sorted correctly

A few it gets wrong. Those are the interesting ones.



AI PROJECT CYCLE: EVALUATION

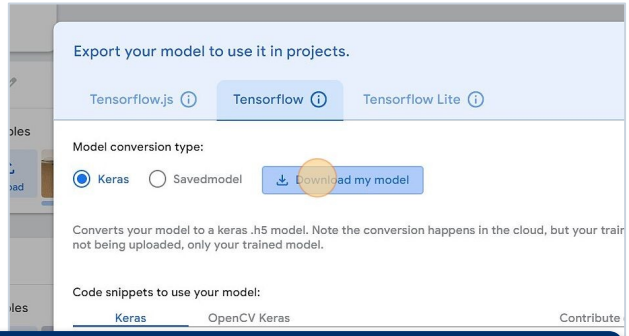
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STEP 6: WATCH IT GO LIVE

I'll run this part. The steps are in the guide for later.

DEMO

Export Model > Upload my model, then paste the link into the Codespaces demo



Take it back to work: with a free GitHub account, anyone on your team can run the same demo with a webcam.

AI PROJECT CYCLE: DEPLOYMENT

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WHAT DID YOUR MODEL LEARN?

Debrief

- ➔ What did it get right, and what did it get wrong?
- ➔ Training data: it only knows the pictures you gave it
- ➔ Overfitting: great on training pictures, worse on new ones?
- ➔ Bias: what if every OK picture were bright and every defect dark?
- ➔ Lighting: what would change on your floor?

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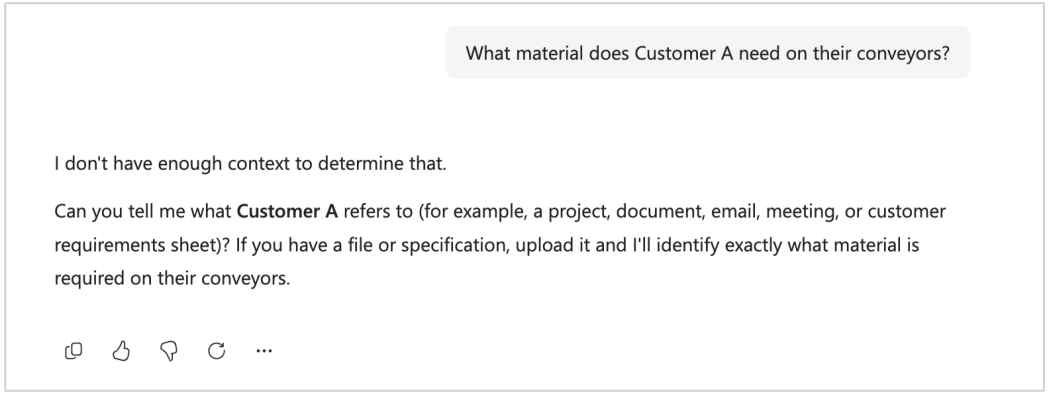
KEY TAKEAWAYS

- ➔ Start with one painful inspection point
- ➔ Your data decides what the model learns
- ➔ Let operations lead, then build confidence with a pilot

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AI FORGETS. EVERY CHAT STARTS OVER.

Ask Copilot about your own customer, and it has no idea who you mean.



Notice what it asks for: a file or a spec. In other words, context.

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CONTEXT BEATS PROMPTS

OPTION A

Describe it

Spend 10 minutes typing out how you like your quotes written.

OPTION B

Show it

Hand Copilot three past quotes and say: “Write this one the same way.”

Outcome + Context is the only framework you need. But good context is wasted if you retype it every time.

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FIND IT. SAVE IT. LET AI KEEP IT.

Good context only pays off if it sticks.

STEP 1

Find it

Outcome + Context, with real examples.

STEP 2

Save it

Put it somewhere that lasts, so you never explain it twice.

STEP 3

Let AI manage it

You set the rules. AI does the upkeep.

WHAT YOU SAVE: THREE PARTS

The same three parts work in Claude and in Copilot.

Instructions Rules that always apply
Claude: CLAUDE.md | Copilot: Customize Copilot, Notebook instructions

Knowledge Reference material
Claude: Resources folder | Copilot: Notebook files

Memory What changes over time
Claude: MEMORY.md | Copilot: Saved memories, memory file

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HOW I RUN MY WORK WITH CLAUDE

Root rules
CLAUDE.md
How I want it to work, always and never.

Root memory
MEMORY.md
Active projects and core facts about me.

Workstations
One per area
Each part of my job, with its own rules and memory.

Resources
Inside each HQ
Every file for that HQ: decks, data, documents.

My root folder CLAUDE.md + MEMORY.md

Canvas HQ	NSF ATE Grant	Department Head HQ	Doctoral Program HQ	Claude SMB HQ	TGW HQ	+ 3 more

Each workstation has its own CLAUDE.md, MEMORY.md, and Resources folder.

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ONE WORKSTATION: NSF ATE GRANT

Where this training lives in my system

- ➔ Instructions: what belongs here and how to work
- ➔ Memory: decisions about this training
- ➔ Resources: the Muskegon decks, datasets, and today's files
- ➔ Rules that learn: “Added after Jon said…”

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WATCH IT REMEMBER

Live in Claude

- ➔ I say “remember this” with a real fact
- ➔ Claude decides where it belongs
- ➔ MEMORY.md updates on its own
- ➔ Next time, it already knows

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WHAT CARRIES OVER TO COPILOT

	Claude	Copilot Basic	Copilot Premium*
Root instructions	Yes	Yes	Yes
“Remember this”	Yes	Yes	Yes
Workstations	Yes	Yes	Yes
Knowledge files	Yes	Yes	Yes
Saves memory by itself	Yes	Partly	Yes
Skills	Yes	No	Yes
Scheduled tasks	Yes	No	Yes
Folders inside folders	Yes	No	No

*Premium: Microsoft 365 Copilot with Copilot Cowork turned on, per Microsoft's documentation. Full table in the guide.

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HOW COPILOT REMEMBERS

1
You say “remember this”

2
Copilot updates memory

3
Copilot saves*

4
It remembers

*Copilot Basic: paste the update into your memory file and upload it again. Premium with Copilot Cowork saves it for you.

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MEET EXAMPLE EQUIPMENT CO.

A fictional company for today's demo

Who

60 people in West Michigan. Custom conveyors for food and industrial plants.

Quoting

Request by email. Engineer scopes it, estimator prices it, quote goes out.

The pain

Customer details live in people's heads and get retyped every quote.

Its HQ

Instructions, a price list, a past quote, and a memory file.

Customers: A, snack food plant. B, pet food plant. C, automotive parts plant.

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OPEN THE GUIDE

Everything you paste today is in the guide, with a copy button.

Scan for today's guide

github.com/grcc-resendiz/ai-mcc

Then find this label under your name



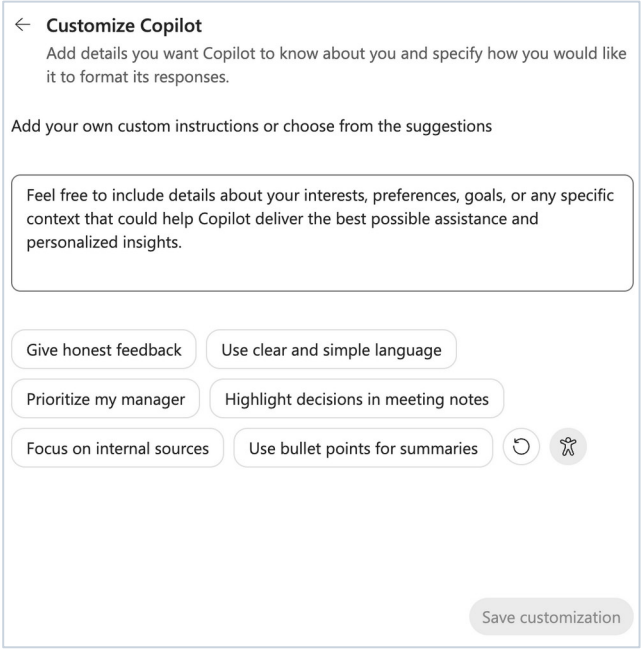
Basic: everything today works for you.
Paid: same steps, plus a few extra features.

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LEVEL 1: LET COPILOT INTERVIEW YOU

1. **Paste the interview prompt from the guide**
2. **Answer 5 quick questions**
3. **Copy the instructions it writes**
4. Gear icon > Personalization > Customize Copilot > Save

ADDS: INSTRUCTIONS



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TEST IT: “REMEMBER THIS”

Your first win

- ➔ Type the first line
- ➔ Look for “Memory saved”
- ➔ Open a new chat, then ask

Look for
Memory saved

ADDS: MEMORY

TYPE	**Remember this: I don't take meetings before my second cup of coffee.**
THEN ASK	**When's a good time to schedule a meeting with me?**

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LEVEL 2: BUILD THE HQ

Look for Notebooks. On some screens it says Workspaces.

1 **Notebooks > New notebook**
Name it Example Equipment Co. HQ

2 **Click ... > Instructions**
Top right of the notebook

3 **Paste the HQ instructions**
From the guide, then Save

ADDS: INSTRUCTIONS

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ADD KNOWLEDGE AND MEMORY

Download three files from the guide, then one upload.

Example Equipment Co.
Price List — KNOWLEDGE

Example Equipment Co.
Past Quote — KNOWLEDGE

Example Equipment Co.
MEMORY — MEMORY

References

- Example Equipment Co. Price List.docx
- Example Equipment Co. Past Quote.docx
- Example Equipment Co. MEMORY.docx

A re-upload may add a number, like “MEMORY 1.”

TRY IT, PROMPT 1: DRAFT A QUOTE

TYPE

Draft a quote for this request in the same format as our past quote:

Then paste Customer A's request from the guide.

EXPECT

Our quote format

Painted steel prices

The 5 percent surcharge from memory

Total of about $35,070

Item	Qty	Unit price	Total
Flat belt conveyor, painted carbon steel frame	40 ft	$450	$18,000
Incline section, any frame (add on)	1	$4,000	$4,000
Motor and gearbox, 1 HP	1	$2,200	$2,200
Safety guarding, painted	40 ft	$125	$5,000
Basic control panel	1	$3,000	$3,000
Engineering and drawings	8 hrs	$150	$1,200
Subtotal			$33,400
Material surcharge (5%)			$1,670
Total			**$35,070**

The surcharge isn't in the past quote. It came from memory.

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PROMPT 2: WHAT'S OPEN

Use it like a coordinator, not just a quote machine.

TYPE

What quotes are open and what's due first?

EXPECT

Customer C, due November 5

Customer A, due November 6

A still shows as open



Copilot

There are **2 open quote requests**:

Customer	Request	Due Date
Customer C	Replacement belt and guarding for an existing line	**November 5, 2026**
Customer A	40 foot incline conveyor for bagged snacks	**November 6, 2026**

Due first: Customer C's quote is due **November 5, 2026**, which is the earliest deadline.

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PROMPT 3: THE GUARDRAIL

TYPE

Customer A also wants a metal detector on this conveyor. Add it to the quote.

EXPECT

"Not on the price list, needs an estimate from engineering"

No made up price

Total stays $35,070

Engineering and drawings	8 hrs	$150	$1,200
Metal detector	1	**Not on the price list, needs an estimate from engineering**	**Not included**
Subtotal (priced items only)			$33,400
Material surcharge (5%)			$1,670
Total (excluding metal detector)			**$35,070**

One line in the instructions: never give a price that isn't on the price list.

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NOW, REMEMBER THIS

TYPE

Remember this: Customer A requires stainless steel frames and guarding on every conveyor in their plant because of daily washdown.

Example Equipment Co. MEMORY
Last updated: November 3, 2026

Customers
Customer A: snack food plant. Buys conveyors for packing lines. Contact is their purchasing manager. Requires stainless steel frames and stainless steel guarding on every conveyor in their plant because of daily washdown.
Customer B: pet food plant. Wants a recommended spare parts kit included in every quote.
Customer C: automotive parts plant. Expects quotes within 3 business days and pays Net 45.

Open Requests
Customer A: 40 foot incline conveyor for bagged snacks, quote due November 6.
Customer C: replacement belt and guarding for an existing line, quote due November 5.

Key Decisions
October 2026: Through December 31, add a 5 percent material surcharge to all quotes because steel prices went up.

EXPECT

The full memory file comes back

New fact on Customer A's line

Nothing else changed

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SAVE IT: COPILOT WRITES, YOU SAVE

1. Click Copy under Copilot's reply
2. Open the MEMORY file in Word
3. Select all, paste, and save
4. In the notebook: ... > Remove the old file
5. Upload the saved file again

Good to know: the upload may show a number, like “MEMORY 1.” That's fine. On Premium with Copilot Cowork, this whole step is automatic.

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ASK AGAIN

Start a new chat inside the notebook, so nothing carries over from before.

TYPE

Redo the Customer A quote.

Item	Qty	Unit Price	Total
Stainless steel conveyor frame	40 ft	$1,000	$40,000
Incline section (add on)	1	$4,000	$4,000
Motor and gearbox, 1 HP, washdown rated	1	$3,500	$3,500
Safety guarding, stainless	40 ft	$250	$10,000
Basic control panel	1	$3,000	$3,000
Engineering and drawings	8 hrs	$150	$1,200
Subtotal			$61,700
Material surcharge (5%)			$3,085
Total			**$64,785**

EXPECT

Stainless frame and guarding

Washdown motor, on its own

About $64,785

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BEFORE AND AFTER

	Before	After
Conveyor frame, 40 ft	$18,000	$40,000
Guarding, 40 ft	$5,000	$10,000
Motor	$2,200	$3,500
Total with surcharge	**$35,070**	**$64,785**

Same request. Same tool. One saved fact nearly doubled the quote.

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YOUR TURN

Let's build one together

- ➔ What would you put in memory for your customers?
- ➔ What rule do your estimators always follow?

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MAKE IT YOURS: FOUR STARTER HQS

Requests and Quotes

Customer requests, estimates, quotes, and follow ups.

Production Data

Line data, shift numbers, graphs, and analysis.

Reports and Communication

Weekly reports, emails, and updates to leadership.

Team

Coaching notes, training plans, and team updates.

All four are in the guide, ready to paste at work.

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WHAT'S NEXT

- ➔ Day 2: turn your HQ into skills and workflows
- ➔ Bring one task you repeat every week
- ➔ Day 3: build AI agents and decide where they help

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